Scenario D: Long-Job Sensitivity Case

Input

Quantum = 3

Process	Arrival Time	Burst Time
P1	0	20
P2	1	3
P3	2	2
P4	3	4

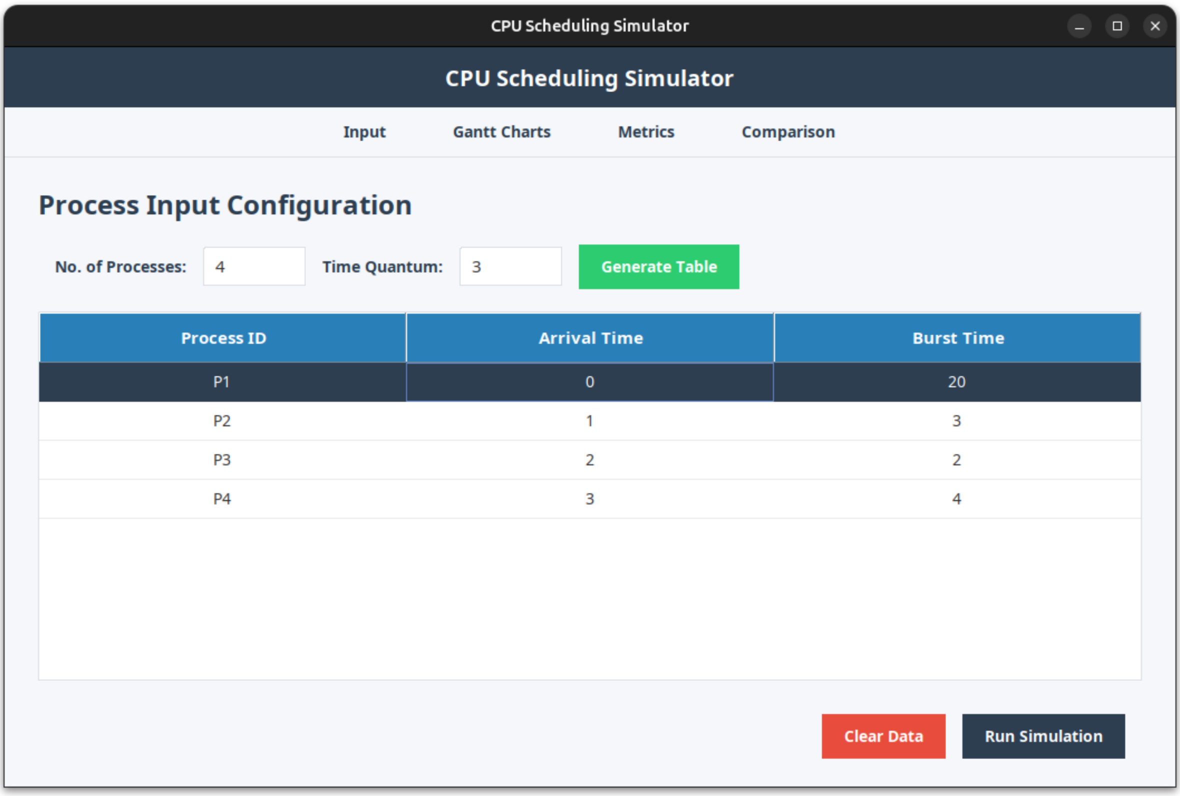
Analysis

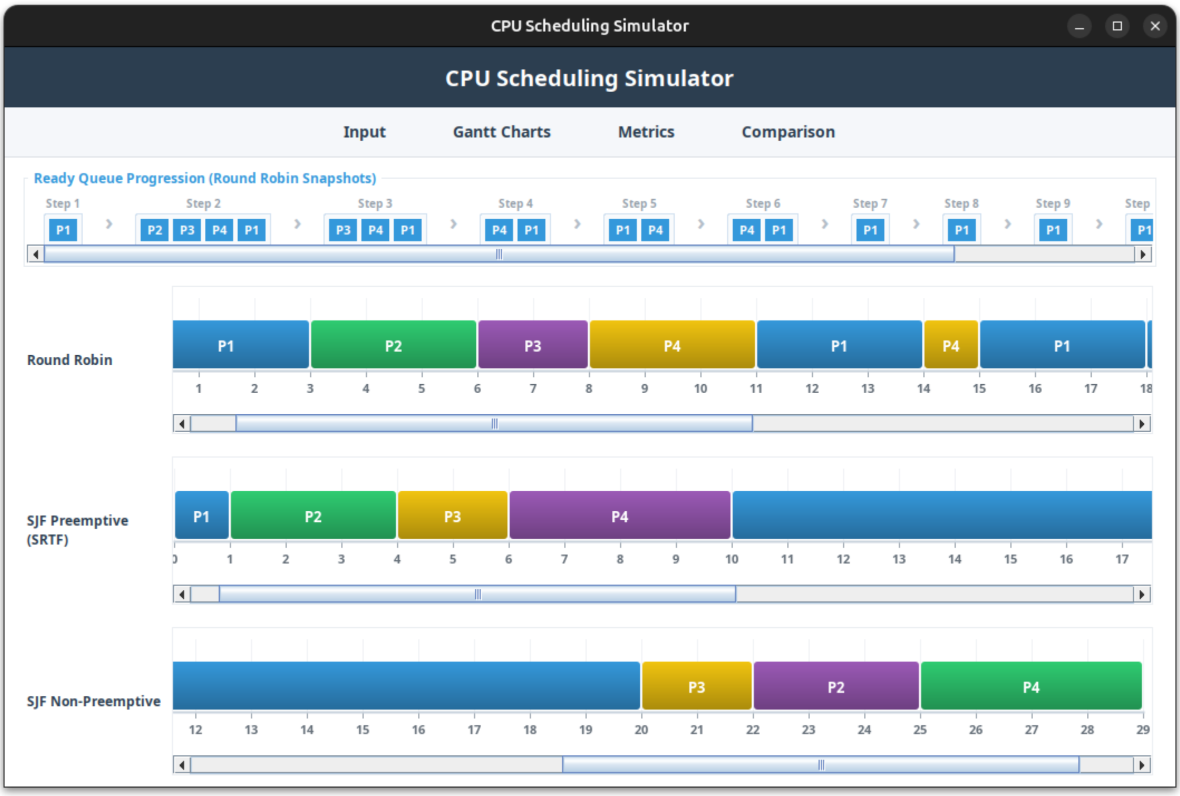
- **Which algorithm gave lower average waiting time?**
  SRTF achieved the lowest average waiting time. Round Robin performed better than SJF.
- **Which algorithm gave lower average response time?**
  SRTF had the best response time because short processes started immediately after arrival.
- **Did Round Robin appear fairer across all processes?**
  Yes. Round Robin ensured that the long process continuously received CPU time and was never ignored.
- **Did SJF complete short jobs more efficiently?**
  SJF completed short jobs efficiently after the long process finished, but SRTF handled them much faster through preemption.
- **How did the chosen quantum affect Round Robin behavior?**
  The quantum allowed all processes to make progress gradually, improving fairness but increasing turnaround time.
- **Which algorithm would you recommend for the tested workload, and why?**
  SRTF is recommended for efficiency, while Round Robin is recommended when fairness and starvation prevention are important.

Conclusion

- SRTF performed best in waiting time, turnaround time, and response time.
- Round Robin was the most balanced and fair algorithm.
- SJF performed poorly because the long process occupied the CPU for a long period before shorter jobs executed.

- The selected quantum in Round Robin prevented starvation and allowed all processes to progress fairly.





CPU Scheduling Simulator

CPU Scheduling Simulator

Input | Gantt Charts | **Metrics** | Comparison

Round Robin Performance

Process ID	Waiting Time (WT)	Turnaround Time (TAT)	Response Time (RT)
P1	9	29	0
P2	2	5	2
P3	4	6	4
P4	8	12	5

Average WT: 5.75 | **Average TAT:** 13.00 | **Average RT:** 2.75

SJF Preemptive (SRTF) Performance

Process ID	Waiting Time (WT)	Turnaround Time (TAT)	Response Time (RT)
P1	9	29	0
P2	0	3	0
P3	2	4	2
P4	3	7	3

Average WT: 3.50 | **Average TAT:** 10.75 | **Average RT:** 1.25

SJF Non-Preemptive Performance

Process ID	Waiting Time (WT)	Turnaround Time (TAT)	Response Time (RT)
P1	0	20	0
P2	21	24	21
P3	18	20	18
P4	22	26	22

Average WT: 15.25 | **Average TAT:** 22.50 | **Average RT:** 15.25

CPU Scheduling Simulator

CPU Scheduling Simulator

Input | Gantt Charts | Metrics | **Comparison**

Comparative Scheduling Analysis

📊 Comparison Analysis

Waiting & Response Time:
- *SJF Preemptive (SRTF)* achieved the **lowest average waiting time**.
- *SJF Preemptive (SRTF)* provided the **fastest response** to new processes.

⚖ **Fairness & Distribution:**
- **Round Robin** is the most fair; waiting times are evenly distributed.

⏱ **Quantum Effect (Q=3):**
- Balanced quantum: Good trade-off between fairness and efficiency.

Final Result:
- Best Overall Waiting Time: **SJF Preemptive**